

InterviewIQ — Project Presentation & Viva Guide

Complete Technical Architecture, Algorithm Reference & Examiner Viva Q&A;

1. Exact Algorithms & Models Reference Table

Module Component	Exact Algorithm / Model Name	Technical Purpose
Adaptive Interview	Item Response Theory (2PL IRT) & Elo Rating Formula	Matches question difficulty (b=1,2,3) dynamically to candidate rating theta
Question Generator	Groq LLaMA-3 8B (llama3-8b-8192)	Sub-200ms real-time question generation & multi-lingual prompt translation
Answer Evaluator	Anthropic Claude 3.5 Sonnet	Scores technical accuracy, depth, clarity & structure (0-100 scale)
Anti-Repetition Engine	Jaccard Similarity Coefficient	Filters duplicate question tokens across sessions ($J(A,B) > 0.55$ threshold)
Scanned Resume OCR	EasyOCR (PyTorch CRNN + CTC Model)	Extracts text from scanned image PDFs via pre-warmed background thread
Face & Eye Detection	Viola-Jones Haar Cascades (OpenCV)	Real-time face & eye ROI tracking (haarcascade_frontalface & eye.xml)
Low-Light Detection	Histogram Equalization & Luminance Thresholding	Equalizes image contrast & flags dim room lighting (Luminance < 20)
Voice Pitch & Tone	pYIN (Probabilistic YIN Algorithm)	Calculates vocal pitch variance F0 to measure monotone vs confident speech
Speech Rate & Fillers	Short-Time Energy (STE) & Google STT WPM Analysis	Detects WPM speed (120-160 WPM optimal) and flags verbal hesitation markers

2. Top 10 Project Guide & External Examiner Viva Questions

Q1: What is the core novelty of InterviewIQ compared to standard online platforms like HackerRank?

■ **Perfect Answer:** InterviewIQ introduces a multi-modal adaptive interview experience. Standard platforms use static pre-defined questions. InterviewIQ dynamically adjusts question difficulty using Item Response Theory (IRT) and Elo Rating, while evaluating candidates across 3 vectors: Technical Concept Depth (Claude 3.5 Sonnet), Facial Proctoring Stability (OpenCV), and Vocal Delivery Confidence (Librosa).

Q2: Why did you use two different LLMs (Groq LLaMA-3 and Anthropic Claude 3.5)?

■ **Perfect Answer:** We designed a hybrid dual-LLM architecture: Groq LLaMA-3 8B provides sub-200ms latency for real-time question generation and prompt translation. Anthropic Claude 3.5 Sonnet is dedicated to deep multi-criteria evaluation of candidate answers with structured JSON scoring.

Q3: How do you prevent asking duplicate or similar questions to the same candidate?

■ **Perfect Answer:** We implemented a canonical anti-repetition engine using Jaccard Similarity ($J(A,B) = |A \cap B| / |A \cup B|$). Generated questions are normalized by removing filler words and stemming tokens. If similarity exceeds 0.55, the prompt is rejected and regenerated. Fingerprints are saved in MongoDB.

Q4: Explain how your Adaptive Engine adjusts question difficulty dynamically.

■ **Perfect Answer:** The Adaptive Engine uses the 2-Parameter Logistic (2PL) Item Response Theory Model and Elo Rating Formula ($R_{\text{new}} = R_{\text{old}} + K*(S-E)$). High technical accuracy increases candidate ability rating θ , generating hard questions ($b=3$). Lower scores adapt difficulty down to solidify foundational concepts.

Q5: How do you parse text from scanned or image-based PDF resumes efficiently?

■ **Perfect Answer:** We use a 3-stage fallback pipeline: pdflumber -> pdfminer -> EasyOCR (CRNN+CTC). To eliminate scanned PDF delays, we pre-warm the EasyOCR neural network model in a background thread at server startup so OCR text extraction completes in under 1 second.

Q6: What algorithms are used for facial proctoring and low-light warnings?

■ **Perfect Answer:** We use OpenCV Multi-Scale Haar Cascade Classifiers. We apply Histogram Equalization (`cv2.equalizeHist`) to boost contrast in dark room lighting, threshold luminance (<20) for low-light alerts, and isolate the upper 65% face ROI to track eye contact (`haarcascade_eye.xml`).

Q7: How is voice confidence measured from microphone input?

■ **Perfect Answer:** We process audio via Librosa: (1) Pitch variance (F_0) is calculated using the pYIN Algorithm to detect monotone vs confident speech, (2) Speech rate is measured in Words Per Minute (120-160 WPM optimal), and (3) Silence pauses are flagged using Short-Time Energy (STE) thresholding.

Q8: How does the system ensure code execution safety in the Coding module?

■ **Perfect Answer:** Candidate code runs inside an isolated subprocess sandbox. We enforce a 3.0-second CPU execution timeout to prevent infinite loops or DoS attacks, while blocking unsafe system calls and capturing stdout, stderr, memory usage, and execution time.

Q9: How does the Proctoring Integrity Monitor handle session termination?

■ **Perfect Answer:** We monitor 4 violation vectors: Tab Switch, Window Blur, Camera Exit / No Face, and Multiple Faces. When a violation occurs, UI alerts display a strike counter. Reaching 3 accumulated strikes automatically terminates the session and flags the recruiter report.

Q10: How do you calculate the final Composite Readiness Index?

■ **Perfect Answer:** We calculate Readiness Index = $(0.50 * \text{Technical Score}) + (0.30 * \text{Voice Confidence}) + (0.20 * \text{Emotion Stability})$. The resulting score maps to 4 readiness badges: Interview Ready (80-100), Improving (60-79), Needs Practice (40-59), and Early Stage (0-39).